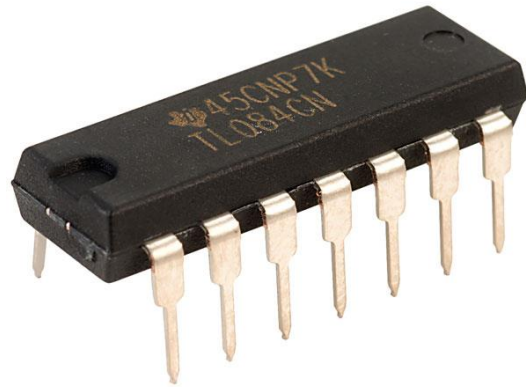
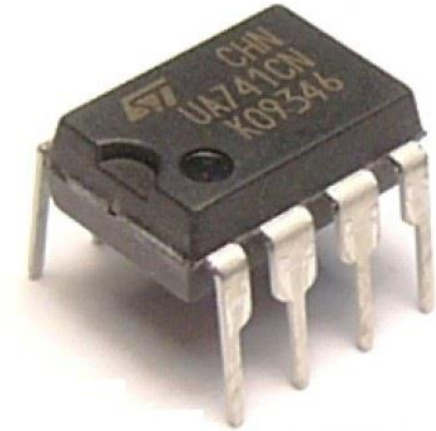
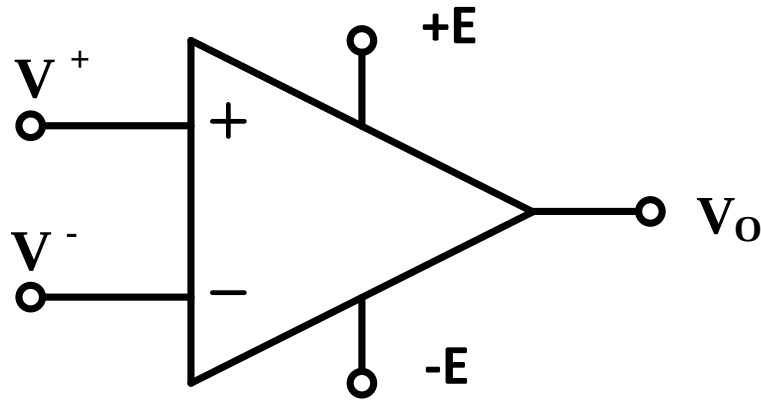


Bazele Circuitelor Electronice

Amplificatoare Operaționale



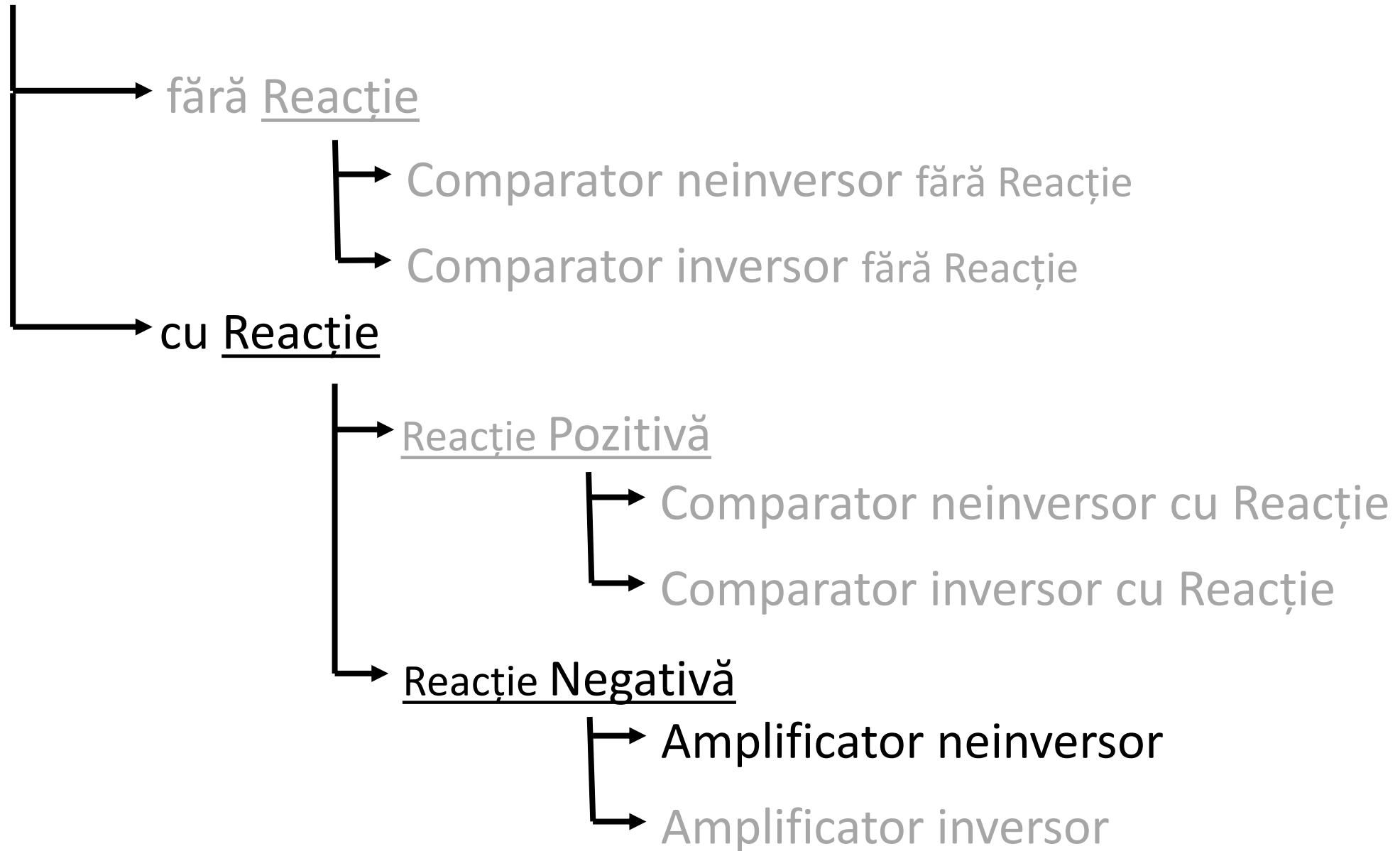
Symbol



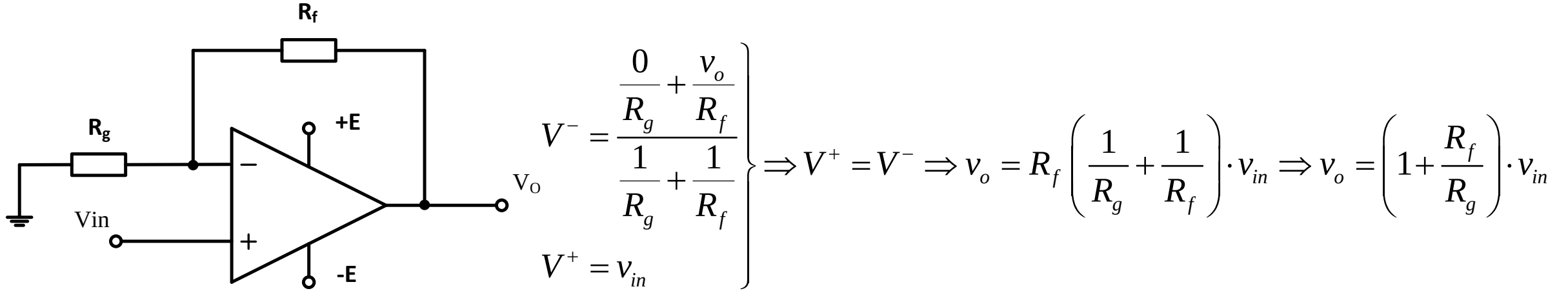
$$v_o = A_D \cdot (v^+ - v^-) = A_D \cdot V_D$$

A_D – amplificare diferențială
în buclă deschisă

Circuite cu AO



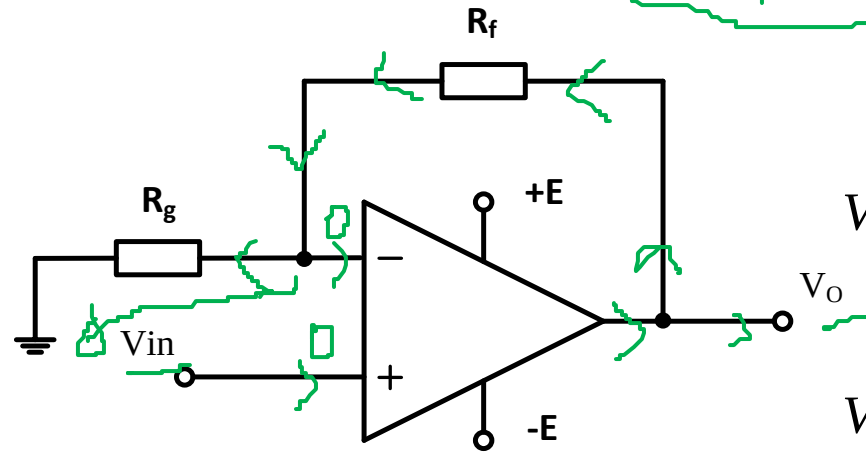
Amplificator neinversor



$$A = \frac{v_o}{v_{in}} = 1 + \frac{R_f}{R_g}$$



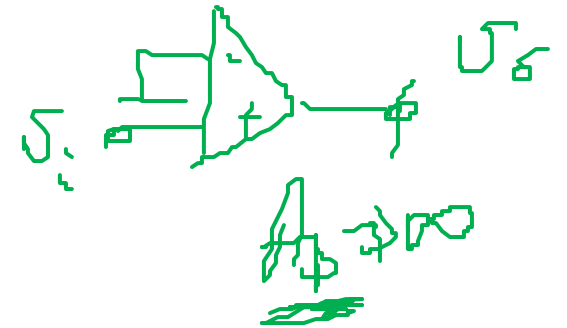
Amplificator neinversor



$$\left. \begin{aligned} V^- &= \frac{\frac{0}{R_g} + \frac{v_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} \\ V^+ &= v_{in} \end{aligned} \right\}$$

$$\Rightarrow V^+ = V^- \Rightarrow v_o = R_f \left(\frac{1}{R_g} + \frac{1}{R_f} \right) \cdot v_{in} \Rightarrow v_o = \left(1 + \frac{R_f}{R_g} \right) \cdot v_{in}$$

$$A = \frac{v_o}{v_{in}} = 1 + \frac{R_f}{R_g}$$

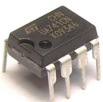


$$v_o = \frac{R_f + R_g}{R_g} v_{in} \Rightarrow v_o = \left(1 + \frac{R_f}{R_g} \right) v_{in}$$

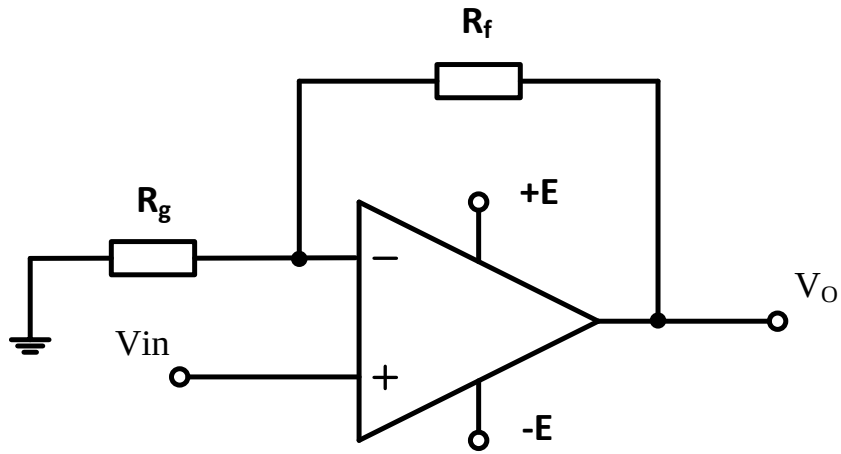
$$v_o = \frac{R_g + R_f}{R_g} v_{in}$$

$$v_o = \left(1 + \frac{R_f}{R_g} \right) v_{in}$$

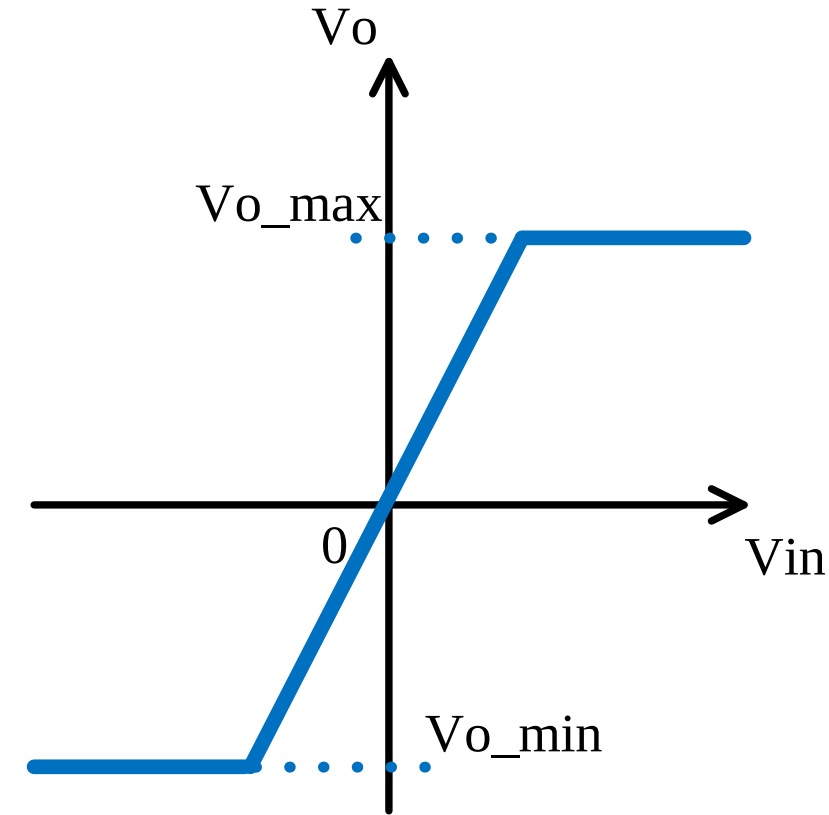
($R_f \rightarrow v_{t2} v_{numa} \cdot v_i = V_p$)



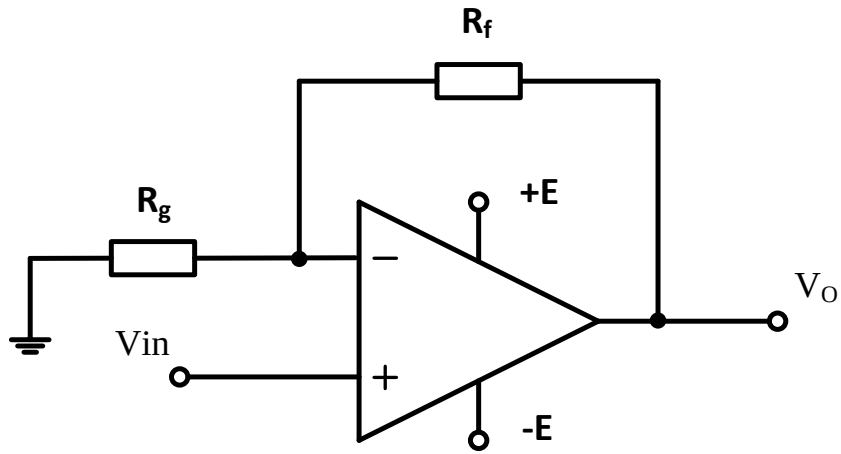
Amplificator neinversor



$$A = \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_g}$$

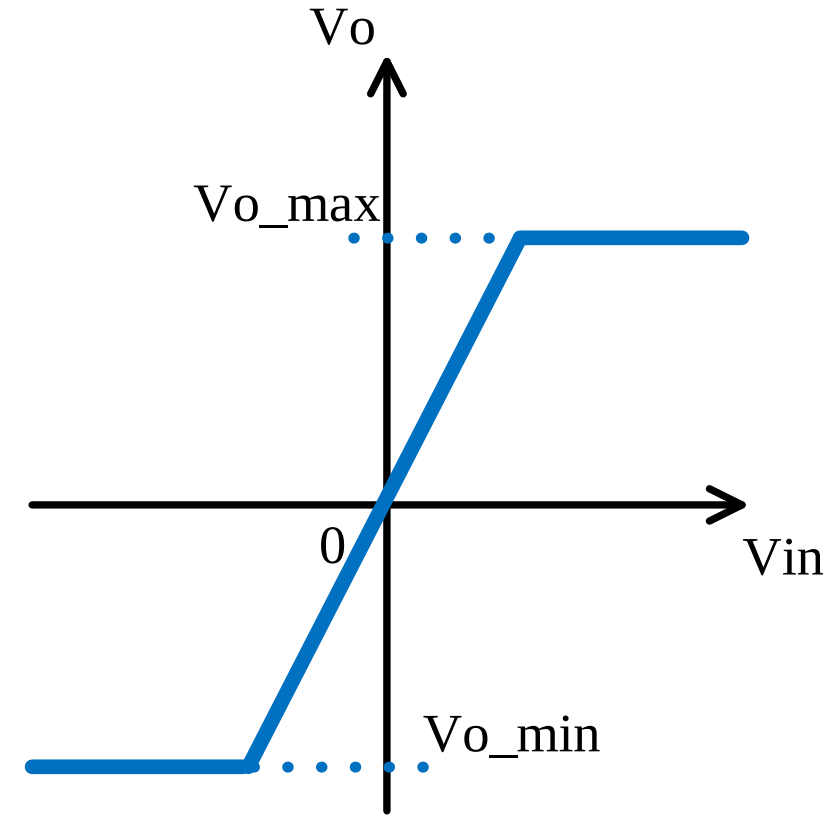


Amplificator neinversor

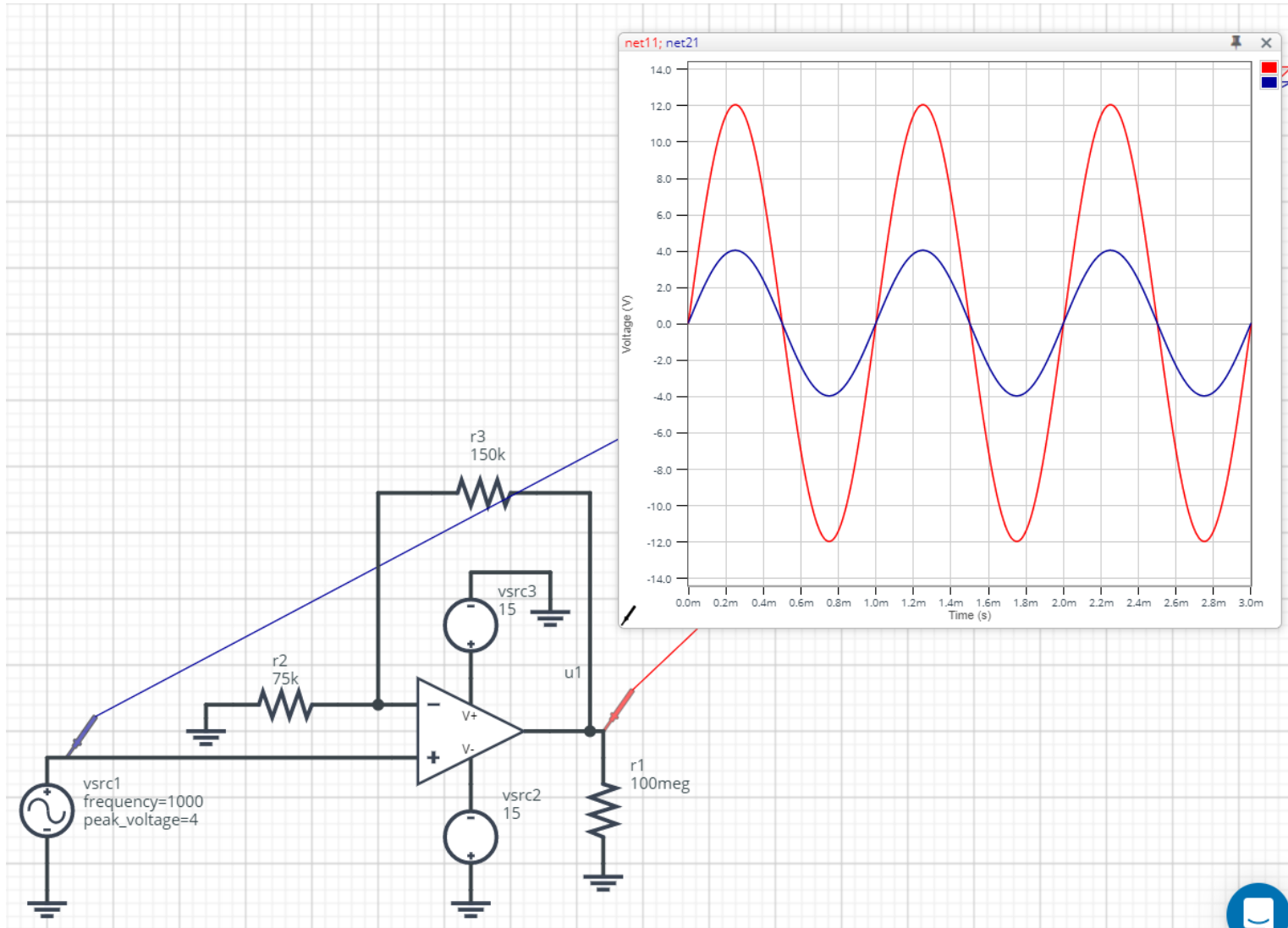


$$A = \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_g}$$

obs : 1. A mai definit de E
2. $A \geq 1$



Amplificator neinversor

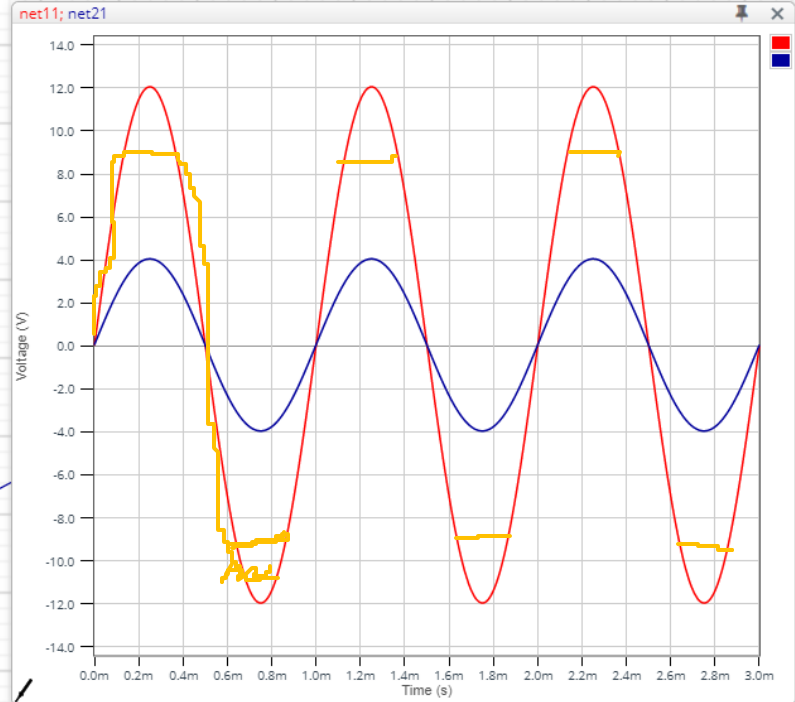


SV C Amplificator neinversor

Amplificator neinversor

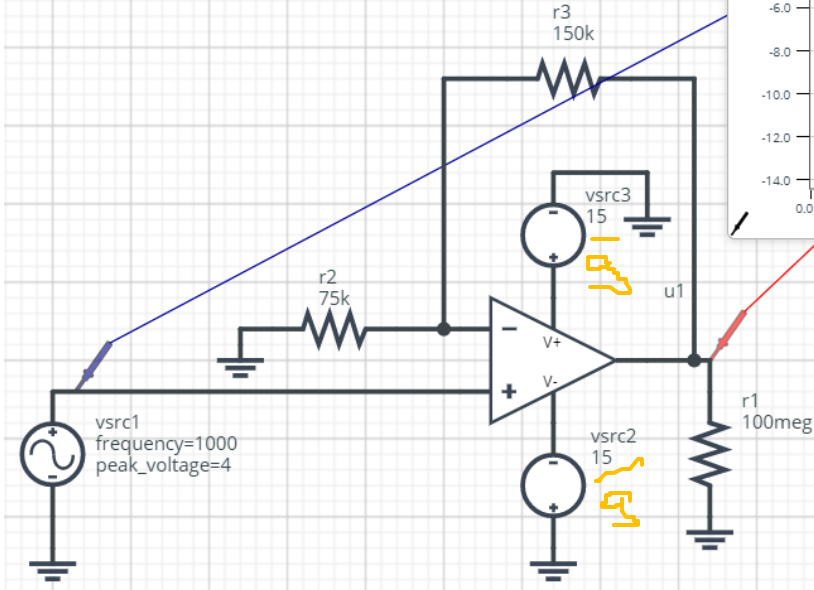
$$R_f = 150k$$

$$R_g = 75k$$



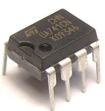
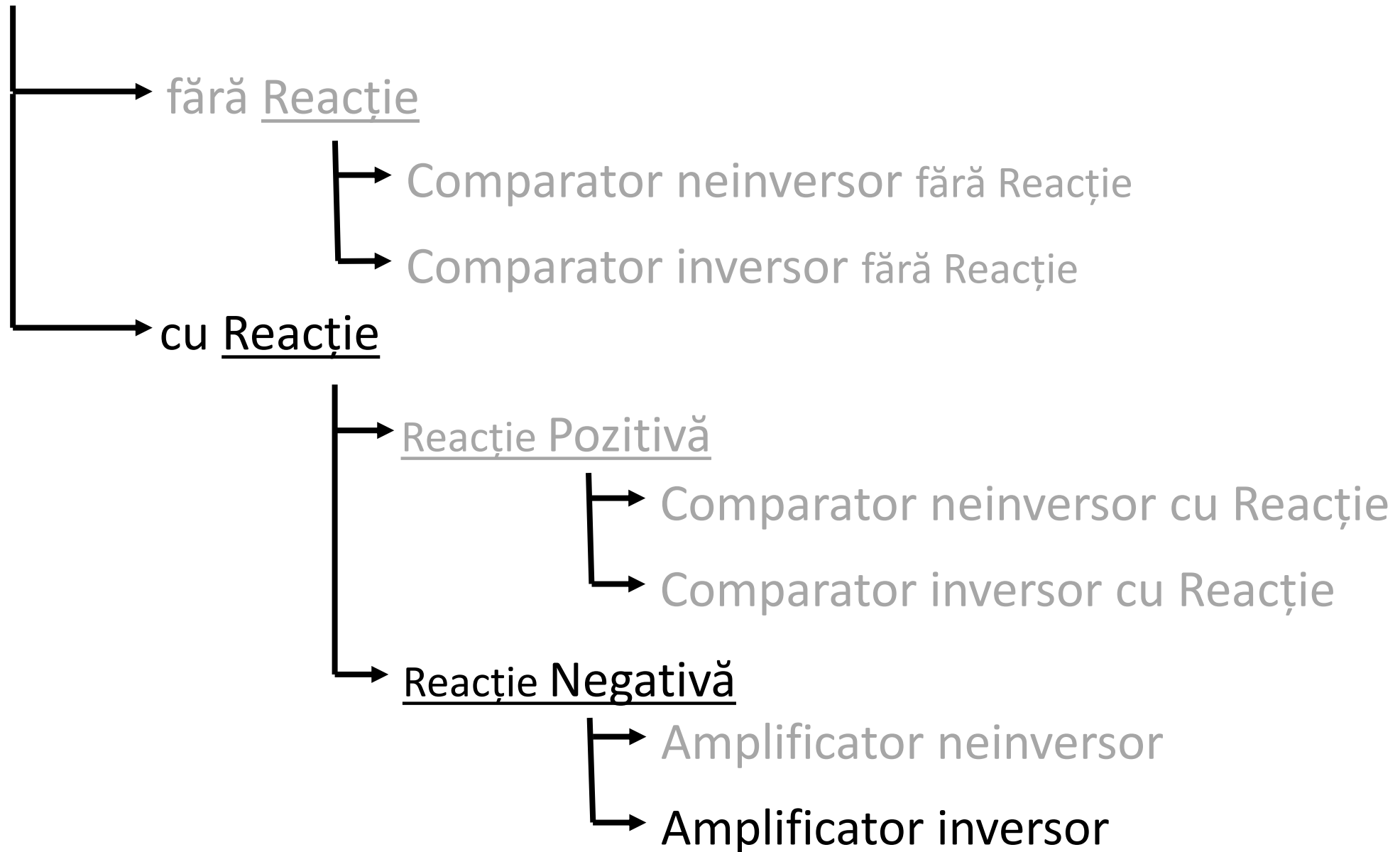
$$A = 1 + \frac{150k}{75k} = 3$$

$$U_{\text{max}} = \pm 9V$$

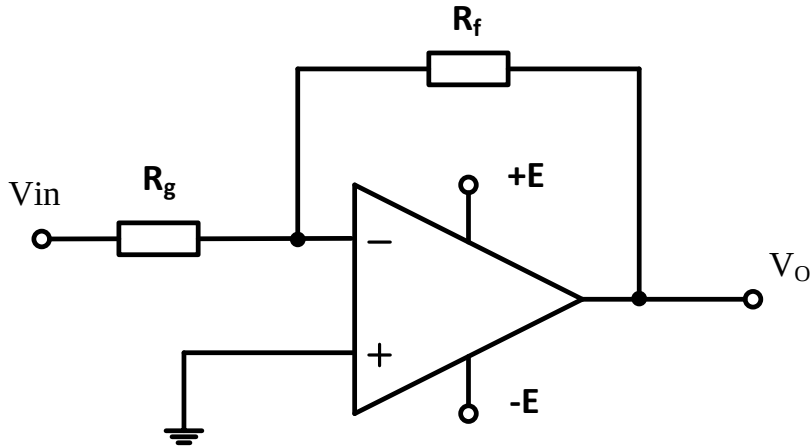


SV C Amplificator neinversor

Circuite cu AO



Amplificator inversor



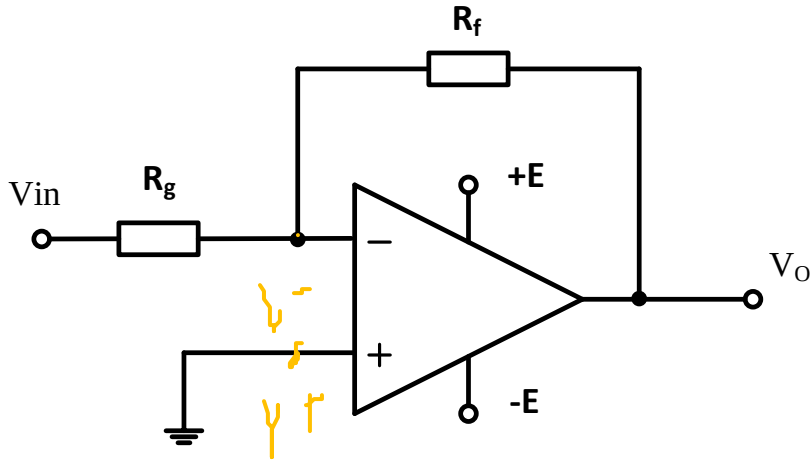
$$\left. \begin{array}{l} V^- = \frac{\frac{V_{in}}{R_g} + \frac{V_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} \\ V^+ = 0 \end{array} \right\} \Rightarrow V^+ = V^- \Rightarrow \frac{\frac{V_{in}}{R_g} + \frac{V_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} = 0$$

$$\Rightarrow \frac{V_{in}}{R_g} + \frac{V_o}{R_f} = 0 \Rightarrow V_o = -\frac{R_f}{R_g} \cdot V_{in}$$

$$A = \frac{V_o}{V_{in}} = -\frac{R_f}{R_g}$$



Amplificator inversor



AN-2 v_f=v₊

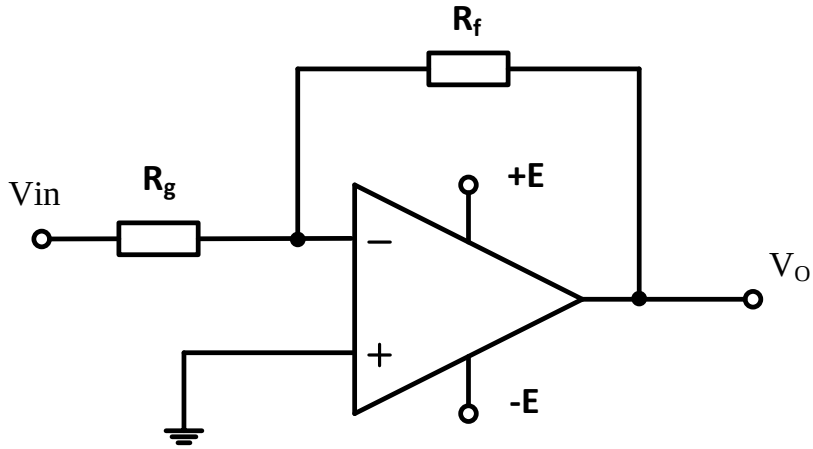
$$V^- = \frac{\frac{V_{in}}{R_g} + \frac{V_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} \quad \Rightarrow \quad V^+ = V^- \Rightarrow \frac{\frac{V_{in}}{R_g} + \frac{V_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} = 0$$

V⁺ = 0

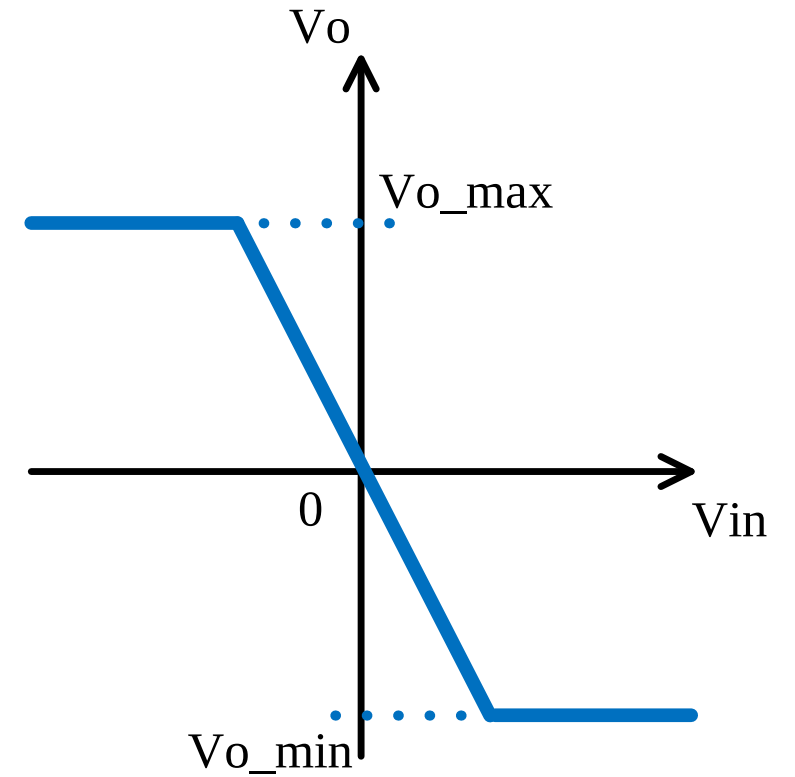
$$\Rightarrow \frac{V_{in}}{R_g} + \frac{V_o}{R_f} = 0 \Rightarrow \underline{V_o} = -\frac{R_f}{R_g} \cdot \underline{V_{in}}$$

$$\underline{A} = \frac{V_o}{V_{in}} = -\frac{R_f}{R_g}$$

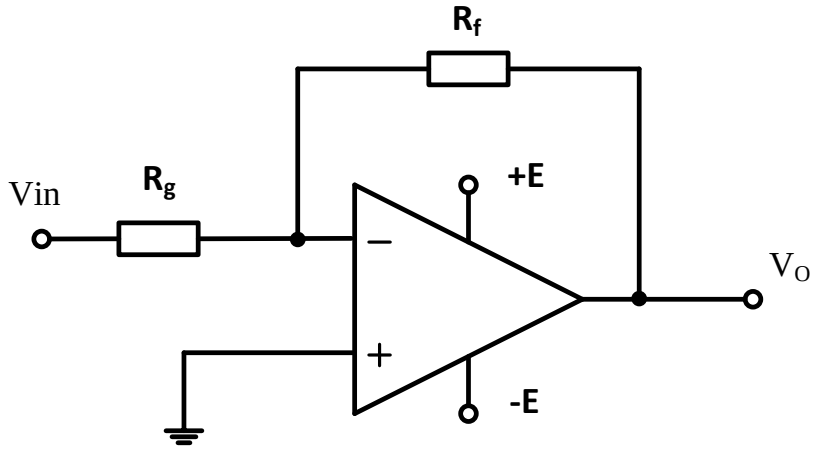
Amplificator inversor



$$A = \frac{v_o}{v_{in}} = -\frac{R_f}{R_g}$$



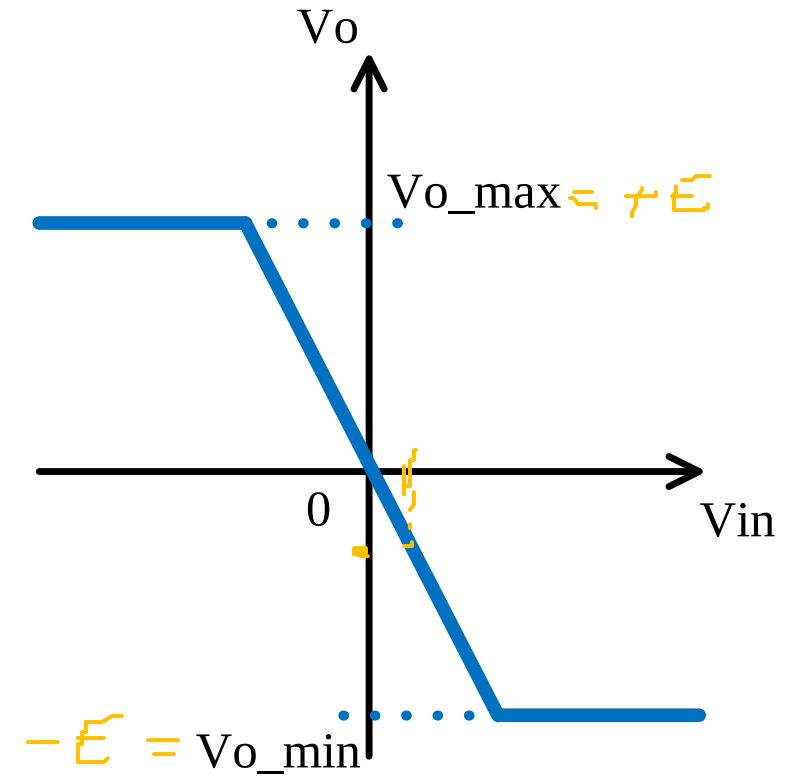
Amplificator inversor



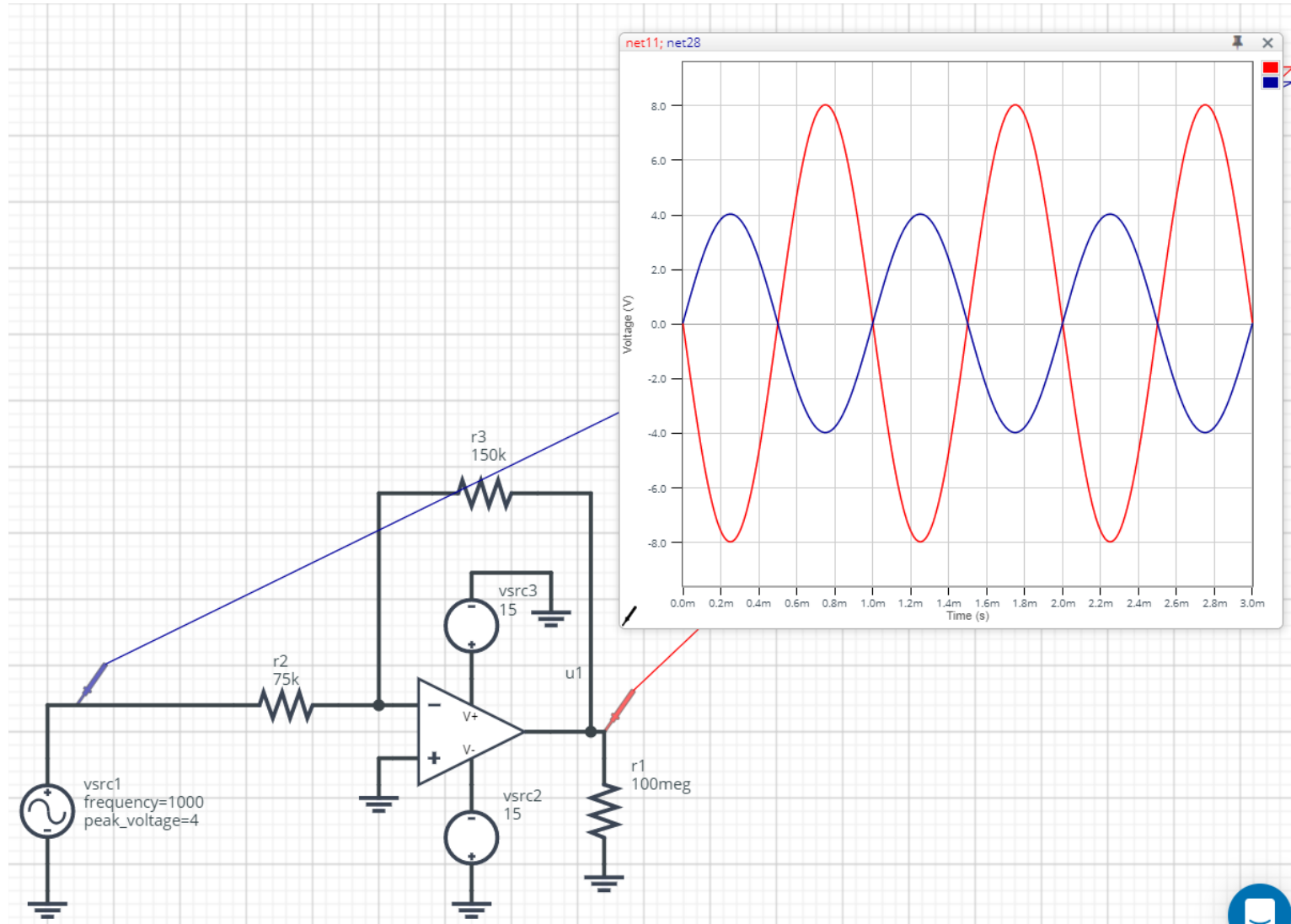
$$A = \frac{V_o}{V_{in}} = -\frac{R_f}{R_g}$$

Obs: 1. Ampla de E

$$2. |A| < 1$$

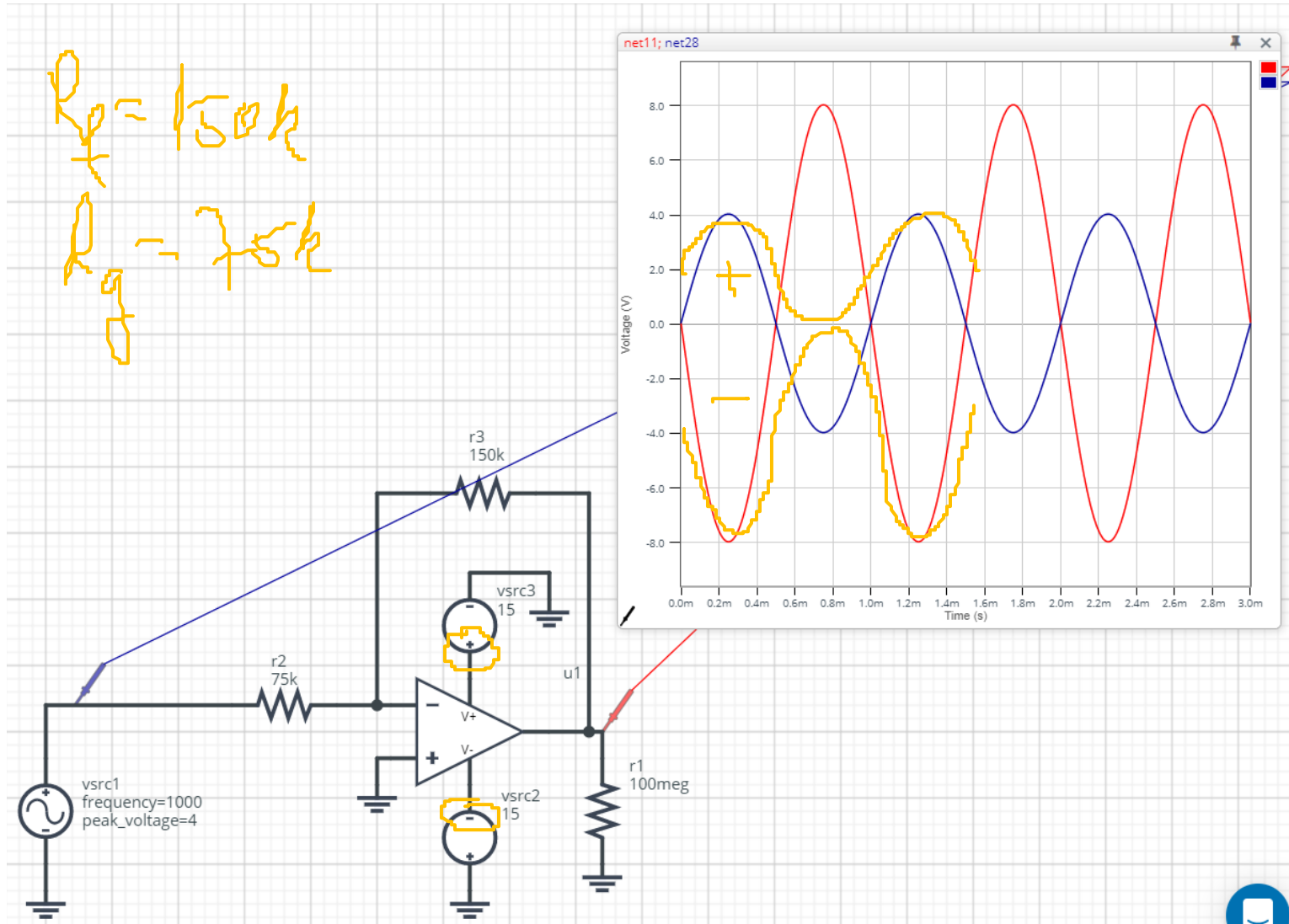


Amplificator inversor



SV C Amplificator inversor

Amplificator inversor



$$A = - \frac{150k}{75k}$$

$$A = -2$$